

A100训练服务器指导手册

1、服务器登录

1.1、事项说明

由于 GPU 卡资源有限，规定一个队伍只能使用一张卡进行训练，否则取消服务器的使用权限。大家可以选择空闲 GPU 卡或者 Volatile GPU-Util 利用率小于 85%的 GPU 卡进行训练，建议优先使用空闲卡进行训练；

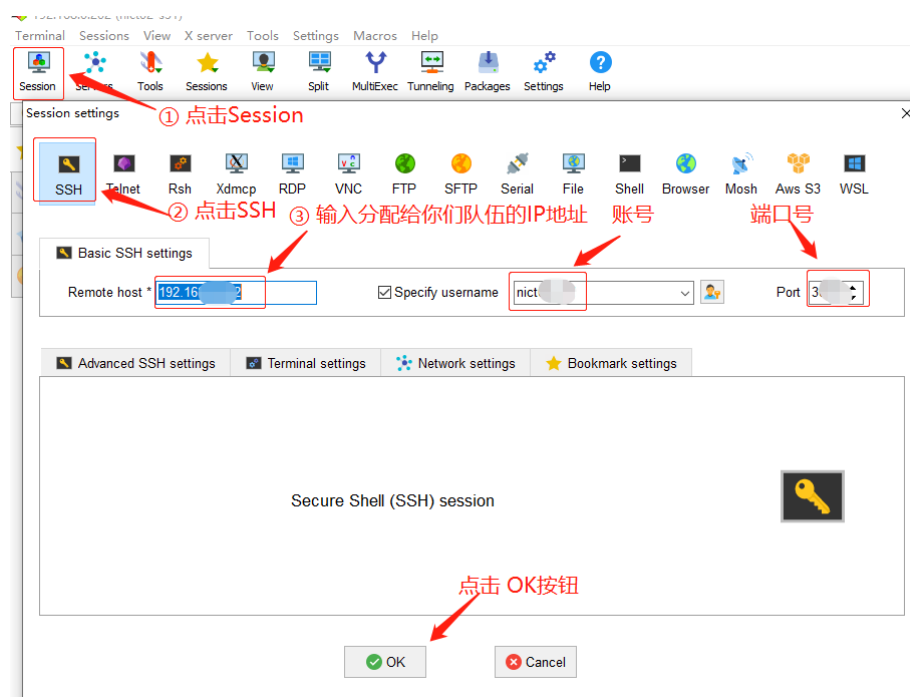
1.2、连接服务前的网络配置

- 方式一：使用SecoClient来进行网络配置
- 方式二：使用Windows自带VPN来进行网络配置

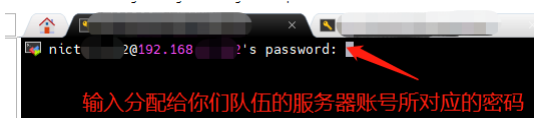
注意：如果以上两种方式，在你和你队友的电脑都无法正常使用，请在QQ群私信 吴老师

1.3、使用SSH登录训练服务器

- 步骤1：下载[MobaXterm工具](#)，当然如果你电脑里面有安装其他的ssh工具也是可以的。
- 步骤2：获取你所在队伍的服务器的IP地址，账号、密码、端口号等，这个可以咨询海思老师或者队长。
- 步骤3：使用MobaXterm登录到，队伍所分配的服务器的ssh。



- 步骤4：输入分配给你们队伍的服务器账号所对应的密码，然后敲回车，就能进入到训练服务器的命令行终端了。



```
• MobaXterm Professional Edition v22.3 •
(SSH client, X server and network tools)

► SSH session to n[redacted]
• Direct SSH : ✓
• SSH compression : ✓
• SSH-browser : ✓
• X11-forwarding : ✗ (disabled or not supported by server)
► For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 5.4.0-137-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

nic@4a43f76a9207:~$
```

- 步骤5: 在训练服务器的命令行终端, 输入下面的命令, 即可看到训练服务器的CUDA版本为11.3

```
nvcc -V
```

```
See 'man sudo_root' for details.
nic@4a43f76a9207:~$ nvcc -V
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2021 NVIDIA Corporation
Built on Mon May 3 19:15:13 PDT 2021
Cuda compilation tools, release 11.3, V11.3.109
Build cuda_11.3.r11.3/compiler.29920130_0
nic@4a43f76a9207:~$
```

1.4、GPU卡使用规则和命名查看

- 用户上线后, 可通过 `nvidia-smi` 查看当前 GPU 资源占用情况, 如下图所示:
- 针对下图 `nvidia-smi` 参数进行解读:
 - GPU: GPU 编号; Name: GPU 型号;
 - Persistence-M: 持续模式的状态。持续模式虽然耗能大, 但是在新的 GPU 应用启动时, 花费的时间更少, 这里显示的是 off 的状态;
 - Fan: 风扇转速, 从 0 到 100%之间变动;
 - Temp: 温度, 单位是摄氏度;
 - Perf: 性能状态, 从 P0 到 P12, P0 表示最大性能, P12 表示状态最小性能 (即 GPU 未工作时为 P0, 达到最大工作限度时为 P12)
 - Pwr:Usage/Cap: 能耗;
 - Memory Usage: 显存使用率;
 - Bus-Id: 涉及 GPU 总线的东西, domain : bus:device.function;
 - Disp.A: Display Active, 表示 GPU 的显示是否初始化;
 - Volatile GPU-Util: 浮动的 GPU 利用率;
 - Uncorr. ECC: Error Correcting Code, 错误检查与纠正;
 - Compute M: compute mode, 计算模式。
 - 下方的 Processes 表示每个进程对 GPU 的显存使用率。
- 注: 若 Volatile GPU-Util 利用率为 0, 则证明该显卡为空闲, 表示 可以使用, 或者某张卡 Volatile GPU-Util 利用率小于 85%也可以使用这张卡, 建议优先使用空闲卡 进行训练。

```
Build cuda_11.3.r11.3/compiler.29920130_0
nict02-s32@4a43f76a9207:~$ nvidia-smi
Fri Feb 10 17:47:32 2023
```

NVIDIA-SMI 515.48.07 Driver Version: 515.48.07 CUDA Version: 11.7									
GPU	Name	Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC			
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util	Compute M.	MIG	M.	
0	Tesla T4	Off	00000000:86:00:0	Off	0%	Default	0	N/A	
N/A	34C	P0	2MiB / 15360MiB						
1	Tesla T4	Off	00000000:87:00:0	Off	0%	Default	0	N/A	
N/A	38C	P0	2MiB / 15360MiB						
2	Tesla T4	Off	00000000:AF:00:0	Off	0%	Default	0	N/A	
N/A	36C	P0	2MiB / 15360MiB						
3	Tesla T4	Off	00000000:D8:00:0	Off	0%	Default	0	N/A	
N/A	38C	P0	2MiB / 15360MiB						


```
Processes:
GPU  GI  CI  PID  Type  Process name  GPU Memory Usage
ID  ID  ID
=====
No running processes found
```

海思目前只提供了基于Resnet18的分类网和YOLOV2的检测网，如果开发者的参赛作品需要用到其他网络的话，需要自己进行训练环境搭建、模型训练、模型转换等工作。

2、分类网模型训练及模型转换

步骤1：数据集准备

- 在进行分类网训练之前，请您仔细阅读社区提供的分类网训练前的准备工作，如：数据集的制作、数据集的清洗、数据集的标注等，具体请参考《[5.3.2.3分类网 \(resent\)](#)》前3小节的内容。

步骤2：数据集上传至训练服务器

- 将制作好的数据集，上传到服务器中，本文提供的垃圾分类案例的数据集已经存放在训练服务器的 `~/ai/sample/trash_classify/dataset` 目录下了。

```
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/dataset$ ls
01_Kitchen_waste_Watermelon_rind  04_Kitchen_waste_Eggplant  07_Hazardous_waste_Waste_battery  10_Hazardous_waste_Medical_gauze  13_Recyclable_waste_Toothbrush
02_Kitchen_waste_Egg_shell        05_Kitchen_waste_Scallion  08_Hazardous_waste_Expired_cosmetics  11_Recyclable_waste_Old_dolls
03_Kitchen_waste_Fishbone         06_Kitchen_waste_Mushroom  09_Hazardous_waste_Woundplast     12_Recyclable_waste_Old_clip
```

步骤3：检查cuda环境

- 查看是否能获取cuda，首先在训练服务器的命令行终端输入 `python`，然后输入 `import torch`，再输入 `torch.cuda.is_available()`，若显示 `true` 则表示cuda 可获取，否则检查环境是否正确，退出输入 `exit()` 即可。

```
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/dataset$ python
Python 3.6.5 [Anaconda, Inc.] (default, Apr 29 2018, 16:14:56)
[GCC 7.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> torch.cuda.is_available()
True
>>> exit() 退出 python环境
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/dataset$
```

步骤4：修改源码和配置文件

- 在训练服务器的命令行终端，执行下面的命令，修改开源代码，需要修改的文件路径在 `~/ai/framework/miclassification/configs/_base_/datasets` 目录下，这里需替换自己的数据集实际路径，按照下图所示进行修改，包括mean、std、resize、crop、data_prefix确认修改，这个务必配置对，否则影响模型训练的效果。

```
cd ~/ai/framework/miclassification/configs/_base_/datasets

sudo chmod 777 *

vim imagenet_bs32.py
```

```
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/datasets$ cd ~/ai/framework/miclassification/configs/_base_/datasets
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/datasets$ sudo chmod 777 *
[sudo] password for nict02-s32:
Sorry, try again.
[sudo] password for nict02-s32: 输入密码
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/datasets$ vim imagenet_bs32.py
```

```
# dataset settings
dataset_type = 'ImageNet'
img_norm_cfg = dict(
    mean=[127.5, 127.5, 127.5], std=[127.5, 127.5, 127.5], to_rgb=True)
train_pipeline = [
    dict(type='LoadImageFromFile'),
    # dict(type='RandomResizedCrop', size=224),
    dict(type='Resize', size=(256, 256)),
    dict(type='CenterCrop', crop_size=224),
    dict(type='RandomFlip', flip_prob=0.5, direction='horizontal'),
    dict(type='Normalize', **img_norm_cfg),
    dict(type='ImageToTensor', keys=['img']),
    dict(type='ToTensor', keys=['gt_label']),
    dict(type='Collect', keys=['img', 'gt_label'])
]
test_pipeline = [
    dict(type='LoadImageFromFile'),
    dict(type='Resize', size=(256, 256)),
    dict(type='CenterCrop', crop_size=224),
    dict(type='Normalize', **img_norm_cfg),
    dict(type='ImageToTensor', keys=['img']),
    dict(type='Collect', keys=['img'])
]
data = dict(
    samples_per_gpu=32,
    workers_per_gpu=2,
    train=dict(
        type=dataset_type,
        data_prefix='/home/nict02-s32/ai/sample/trash_classify/dataset',
        pipeline=train_pipeline),
    val=dict(
        type=dataset_type,
        data_prefix='/home/nict02-s32/ai/sample/trash_classify/dataset',
        # ann_file='data/imagenet/meta/val.txt',
        pipeline=test_pipeline),
    test=dict(
        # replace `data/val` with `data/test` for standard test
        type=dataset_type,
        data_prefix='/home/nict02-s32/ai/sample/trash_classify/dataset',
        # ann_file='data/imagenet/meta/val.txt',
        pipeline=test_pipeline))
evaluation = dict(interval=1, metric='accuracy')
```

需要训练的数据集的绝对路径

- 在训练服务器的命令行终端，执行下面的命令，修改 `~/ai/framework/miclassification/configs/_base_/models` 目录下resnet18.py，按下图修改即可：

```
cd ../models

sudo chmod 777 *

vim resnet18.py
```

```
nict02-s32@4a43f76a9207:~/ai/framework/mmlclassification/configs/_base_/datasets$ cd ../models
nict02-s32@4a43f76a9207:~/ai/framework/mmlclassification/configs/_base_/datasets$ sudo chmod 777 *
nict02-s32@4a43f76a9207:~/ai/framework/mmlclassification/configs/_base_/models$ vim resnet18.py
nict02-s32@4a43f76a9207:~/ai/framework/mmlclassification/configs/_base_/models$
```

```
# model settings
model = dict(
    type='ImageClassifier',
    backbone=dict(
        type='ResNet',
        depth=18,
        num_stages=4,
        out_indices=(3, ),
        style='pytorch'),
    neck=dict(type='GlobalAveragePooling'),
    head=dict(
        type='LinearClsHead',
        num_classes=13, ① num_classes根据实际训练数据集的种类来填写
        in_channels=512,
        loss=dict(type='CrossEntropyLoss', loss_weight=1.0),
        topk=(1, 5), ② 当num_classes<5时, topk的第二个值=num_classes
                    当num_classes>=5时, topk的第二个值为5
    ))
```

步骤5：查看显卡资源

- 使命令 `nvidia-smi` 查看哪张显卡空闲或 Volatile GPU-Util 利用率较低，假如第0张显卡空闲，我们就进入到下一步骤。

```

Fri Feb 10 18:51:08 2023
$ nvidia-smi
+-----+
| NVIDIA-SMI 515.48.07    Driver Version: 515.48.07    CUDA Version: 11.7    |
+-----+-----+-----+-----+-----+-----+
| GPU  Name          Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf    Pwr:Usage/Cap|  Memory-Usage | GPU-Util  Compute M. |
|====+=====+====+=====+=====+=====+
|  0   Tesla T4              Off   | 00000000:86:00:0 Off |             0         |
| N/A   35C    P0      28W / 70W   | 1371MiB / 15360MiB | 0%        Default   |
+-----+-----+-----+-----+-----+-----+
|  1   Tesla T4              Off   | 00000000:87:00:0 Off |             0         |
| N/A   38C    P0      27W / 70W   |  2MiB / 15360MiB | 0%        Default   |
+-----+-----+-----+-----+-----+-----+
|  2   Tesla T4              Off   | 00000000:AF:00:0 Off |             0         |
| N/A   36C    P0      27W / 70W   |  2MiB / 15360MiB | 0%        Default   |
+-----+-----+-----+-----+-----+-----+
|  3   Tesla T4              Off   | 00000000:D8:00:0 Off |             0         |
| N/A   38C    P0      28W / 70W   |  2MiB / 15360MiB | 0%        Default   |
+-----+-----+-----+-----+-----+-----+
```

步骤6：开始训练

- 切换到mmlclassification目录下，在训练服务器的命令行终端，执行下面的命令，开始进行模型的训练

```
# 进入到mmclassification目录下
cd ~/ai/framework/mmclassification/

# 开始进行模型训练
CUDA_VISIBLE_DEVICES=0 bash ./tools/dist_train.sh
configs/resnet/resnet18_b32x8_imagenet.py 1 --work-dir ./ckpt

# 对上述命令阐述如下:
# CUDA_VISIBLE_DEVICES=0 表示使用GPU编号为0的显卡进行训练
# dist_train.sh - 训练sh脚本
# configs/resnet/resnet18_b32x8_imagenet.py - 训练依赖的配置
# 1 - GPU个数
# --work-dir ./ckpt - 模型存放的路径
```

```
nict02-s32@4a43f76a9207:~/ai/framework/mmclassification/configs/_base_/datasets$ cd ~/ai/framework/mmclassification/
nict02-s32@4a43f76a9207:~/ai/framework/mmclassification$ CUDA_VISIBLE_DEVICES=0 bash ./tools/dist_train.sh configs/resnet/resnet18_b32x8_imagenet.py 1 --work-dir ./ckpt
/usr/local/anaconda3/lib/python3.6/site-packages/torch/distributed/launch.py:186: FutureWarning: The module torch.distributed.launch is deprecated
and will be removed in future. Use torchrun.
Note that --use_env is set by default in torchrun.
If your script expects --local_rank argument to be set, please
change it to read from os.environ['LOCAL_RANK'] instead. See
https://pytorch.org/docs/stable/distributed.html#launch-utility for
further instructions
FutureWarning:
fatal: not a git repository (or any parent up to mount point /home)
Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).
2023-02-10 18:50:53.205 - mmcls - INFO - Environment Info:
```

- 当开始训练之后，我们可以看到显卡0的Volatile GPU-Util 利用率就已经大于0%了，代表该显卡正在使用中。

```

~\nicu2023$ nvidia-smi
Fri Feb 10 18:54:42 2023
+-----+
| NVIDIA-SMI 515.48.07   | Driver Version: 515.48.07   | CUDA Version: 11.7   |
+-----+
| GPU  Name      Persistence-M | Bus-Id      Disp.A     Volatile Uncorr. ECC | |
| Fan  Temp      Perf      Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|                               |                   | MIG M. |
+-----+
|  0  Tesla T4    Off          | 00000000:86:00:0 Off      |    27%    Default  0 |
| N/A   51C       P0          | 2515MiB / 15360MiB |          | N/A   |
+-----+
|  1  Tesla T4    Off          | 00000000:87:00:0 Off      |     0%    Default  0 |
| N/A   33C       P8          |  2MiB / 15360MiB |          | N/A   |
+-----+
|  2  Tesla T4    Off          | 00000000:AF:00:0 Off      |     0%    Default  0 |
| N/A   31C       P8          |  2MiB / 15360MiB |          | N/A   |
+-----+
|  3  Tesla T4    Off          | 00000000:D8:00:0 Off      |     0%    Default  0 |
| N/A   33C       P8          |  2MiB / 15360MiB |          | N/A   |
+-----+

```

- 注:训练时务必指定某张空闲显卡或 Volatile GPU-Util 利用率较低的显卡进行训练。
- 训练成功后，即可在 ~/ai/framework/mmclassification/ckpt 目录下，查看生成的.pth 模型，其中最后一个 epoch_100.pth是我们所需要的。


```

-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 19:20 epoch_8.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:41 epoch_80.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:44 epoch_81.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:47 epoch_82.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:49 epoch_83.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:52 epoch_84.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:55 epoch_85.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 22:58 epoch_86.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:01 epoch_87.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:03 epoch_88.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:06 epoch_89.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 19:23 epoch_9.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:09 epoch_90.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:12 epoch_91.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:14 epoch_92.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:17 epoch_93.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:20 epoch_94.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:23 epoch_95.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:26 epoch_96.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:28 epoch_97.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:31 epoch_98.pth
-rw-rw-r-- 1 nict02-s32 nict02-s32 86M Feb 10 23:34 epoch_99.pth
lrwxrwxrwx 1 nict02-s32 nict02-s32 13 Feb 10 23:37 latest.pth -> epoch_100.pth
nict02-s32@4a43f76a9207:~/ai/framework/miclassification/ckpt$

```

步骤7：模型转换

- 目前 NNIE 配套软件及工具链仅支持以 Caffe1.0 框架，使用其他框架的网络模型需要转化为Caffe 框架下的模型，具体转换过程如下。
- 在训练服务器的命令行终端，执行下面的命令，对
~/pytorch_to_caffe_master/example/resnet_pytorch_2_caffe.py进行修改，主要需要修改的是 name, num_classes的种类，以及您pytorch模型的绝对路径，具体修改如下图所示，**注意：因为 latest.pth只是epoch_100.pth的一个软连接，我们最后指定epoch_100.pth的绝对路径就好。**

```
cd ~/ai/pytorch_to_caffe_master
```

```
vim example/resnet_pytorch_2_caffe.py
```

```

nict02-s32@4a43f76a9207:~$
nict02-s32@4a43f76a9207:~$
nict02-s32@4a43f76a9207:~$ cd ~/ai/pytorch_to_caffe_master/
nict02-s32@4a43f76a9207:~/ai/pytorch_to_caffe_master$
nict02-s32@4a43f76a9207:~/ai/pytorch_to_caffe_master$ vim example/resnet_pytorch_2_caffe.py

```

```

import sys
sys.path.insert(0, '.')
import torch
from torch.autograd import Variable
from torchvision.models import resnet
import pytorch_to_caffe

if __name__ == '__main__':
    name='resnet18'
    resnet18=resnet.resnet18(num_classes=13)
    checkpoint = torch.load("/home/nict02-s32/ai/framework/miclassification/ckpt/epoch_100.pth", map_location = 'cpu')
    new_sd = {}
    for k,v in checkpoint['state_dict'].items():
        if not k.endswith('num_batches_tracked'):
            if k.startswith('backbone'):
                k_new = k.split('backbone.')[1]
            if k.startswith('head'):
                k_new = k.split('head.')[1]
            new_sd[k_new] = v

    resnet18.load_state_dict(new_sd)
    resnet18.eval()
    input=torch.ones([1,3,224,224])
    #input=torch.ones([1,3,224,224])
    pytorch_to_caffe.trans_net(resnet18,input,name)
    pytorch_to_caffe.save_prototxt('{}{}_prototxt'.format(name))
    pytorch_to_caffe.save_caffemodel('{}{}_caffemodel'.format(name))

```

- 在训练服务器的命令行终端，执行下面的命令,进行模型的转换：

```
python example/resnet_pytorch_2_caffe.py
```

```
nict02-s32@4a43f76a9207:~/ai/pytorch_to_caffe_master$ python example/resnet_pytorch_2_caffe.py
Starting Transform, This will take a while
139672012504184:blob1 was added to blobs
torch ops name: {ResNet(
  (conv1): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3, 3), bias=False)
  (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (relu): ReLU(inplace=True)
  (maxpool): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1, ceil_mode=False)
  (layer1): Sequential(
    (0): BasicBlock(
      (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
    (1): BasicBlock(
      (conv1): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
      (relu): ReLU(inplace=True)
      (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    )
  )
  (layer2): Sequential(
    (0): BasicBlock(
```

- 转换成功后，即可在pytorch_to_caffe_master目录下生成resnet18.caffemodel和resnet18.prototxt文件。

```
conv20 was added to layers
139672012537272:conv_blob20 was added to blobs
['relu_blob16']
layer4.1.bn2
batch_norm20 was added to layers
139672012537928:batch_norm_blob20 was added to blobs
['conv_blob20']
bn_scale20 was added to layers
['batch_norm_blob20']
add8 was added to layers
139672012538328:add_blob8 was added to blobs
['batch_norm_blob20', 'relu_blob15']
layer4.1.relu
relu17 was added to layers
139672012538248:relu_blob17 was added to blobs
['add_blob8']
avgpool
ave_pool1 was added to layers
139672012538168:ave_pool_blob1 was added to blobs
['relu_blob17']
fc
fc1 was added to layers
139672012538568:fc_blob1 was added to blobs
WARNING: CANNOT FOUND blob 139672012538008
Transform Completed
nict02-s32@4a43f76a9207:~/ai/pytorch_to_caffe_master$ ls
001763.jpg  LICENSE.md  README.md  __pycache__  caffe_analyser.py  funcs.py  pytorch_analyser.py  resnet18.caffemodel  visionmaster
Caffe       Pytorch     __init__.py  analysis     example         model      pytorch_to_caffe.py  resnet18.prototxt
nict02-s32@4a43f76a9207:~/ai/pytorch_to_caffe_master$
```

- 至此，模型转换已经完成。

步骤8: prototxt网络层适配

- 关于prototxt网络层适配的相关操作，可以直接参考社区提供的[5.3.2.3.6 prototxt网络层适配](#)小节的内容即可。



步骤9: 模型量化

- 关于模型量化的相关操作，可以直接参考社区提供的 [5.3.2.3.7 模型量化](#) 小节的内容即可。

- 5.3.2.3.3、...
- 5.3.2.3.4 分...
- 5.3.2.3.4...
- 5.3.2.3.4...
- 5.3.2.3.5 Py...
- 5.3.2.3.6 pr...
- 5.3.2.3.6...
- 5.3.2.3.6...
- 5.3.2.3.7 ...**
- 5.3.2.3.8、...
- 5.3.2.3.8...

5.3.2.3.7 模型量化

定义：量化的过程将caffe模型转换成可以在海思芯片上执行的wk文件，同时将fl

相关nnie_mapper配置概念请阅读[源码的device/soc/hisilicon/hi3516dv300/sdk_](#)配置文件说明，如下图所示：

- 3 NNIE开发指南
 - 3.1 NNIE介绍
 - 3.2 Prototxt要求
 - 3.2.1 deploy.prototxt输入层格式
 - 3.2.2 prototxt layer描述格式

步骤10：模型部署板端和调试

- 关于模型部署板端和调试的相关操作，可以直接参考社区提供的 [5.3.2.3.8、模型部署板端和调试](#) 小节的内容即可。

- 5.3.2.3.5 Py...
- 5.3.2.3.6 pr...
- 5.3.2.3.6...
- 5.3.2.3.6...
- 5.3.2.3.7 ...**
- 5.3.2.3.8、...**
- 5.3.2.3.8...

5.3.2.3.8、模型部署板端和调试

5.3.2.3.8.1、概述

- wk模型生成之后，需要将生成的wk文件部署到板端，z
示例代码参考在device/soc/hisilicon/hi3516dv300/sdk_
示：

3、检测网模型训练及模型转换

步骤1：数据集准备

- 在进行检测网训练之前，请您仔细阅读社区提供的检测网训练前的准备工作，如：数据集的制作、数据集的清洗、数据集的标注等，具体请参考《[5.3.2.4检测网](#)》第1小节的内容。

步骤2：数据集上传至训练服务器

- 将制作好的数据集，上传到服务器中，本文提供的手势检测案例的数据集已经存放在训练服务器的~/ai/sample/hand_detect 目录下了。

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$ ls
test_dataset training_dataset
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$
```

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$ ls
test_dataset training_dataset
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect$ cd test_dataset/
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset$ ls
JPEGImages labels list
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset$ cd ../training_dataset/
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset$ ls
JPEGImages labels list
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset$
```

- 接下来对几个文件夹，做一下简要说明。

```
# test_dataset为测试数据
# training_dataset为训练数据
# JPEGImages目录下是需要进行训练的数据集图片
# labels目录下的txt是JPEGImages目录下对应图片所标注的labels
# list目录下的hand_train.txt文件中保存的是JPEGImages目录下所有图片的绝对路径。
# 注意：所有的txt文件都必须是linux格式的，可以使用 dos2unix 工具进行文件格式的转换
# 使用方法： dos2unix filename （如果还不知道如何使用，可上网咨询度娘）
```

- JPEGImages目录下的图片，就是需要进行训练的数据集图片

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/JPEGImages$ ls
Buffy_1.jpg    Buffy_432.jpg    Inria_27.jpg    Inria_98.jpg    Poselet_8.jpg    VOC2007_140.jpg    VOC2007_474.jpg    VOC2007_807.
Buffy_10.jpg   Buffy_433.jpg    Inria_270.jpg   Inria_99.jpg    Poselet_80.jpg   VOC2007_141.jpg   VOC2007_475.jpg   VOC2007_808.
Buffy_100.jpg  Buffy_434.jpg    Inria_271.jpg   Poselet_1.jpg   Poselet_81.jpg   VOC2007_142.jpg   VOC2007_476.jpg   VOC2007_809.
Buffy_101.jpg  Buffy_435.jpg    Inria_272.jpg   Poselet_10.jpg  Poselet_82.jpg   VOC2007_143.jpg   VOC2007_477.jpg   VOC2007_81.
Buffy_102.jpg  Buffy_436.jpg    Inria_273.jpg   Poselet_100.jpg Poselet_83.jpg   VOC2007_144.jpg   VOC2007_478.jpg   VOC2007_810.
Buffy_103.jpg  Buffy_437.jpg    Inria_274.jpg   Poselet_101.jpg Poselet_84.jpg   VOC2007_145.jpg   VOC2007_479.jpg   VOC2007_811.
Buffy_104.jpg  Buffy_438.jpg    Inria_275.jpg   Poselet_102.jpg Poselet_85.jpg   VOC2007_146.jpg   VOC2007_48.jpg    VOC2007_812.
Buffy_105.jpg  Buffy_439.jpg    Inria_276.jpg   Poselet_103.jpg Poselet_86.jpg   VOC2007_147.jpg   VOC2007_480.jpg   VOC2007_813.
Buffy_106.jpg  Buffy_44.jpg     Inria_277.jpg   Poselet_104.jpg Poselet_87.jpg   VOC2007_148.jpg   VOC2007_481.jpg   VOC2007_814.
Buffy_107.jpg  Buffy_440.jpg    Inria_278.jpg   Poselet_105.jpg Poselet_88.jpg   VOC2007_149.jpg   VOC2007_482.jpg   VOC2007_815.
Buffy_108.jpg  Buffy_441.jpg    Inria_279.jpg   Poselet_106.jpg Poselet_89.jpg   VOC2007_15.jpg    VOC2007_483.jpg   VOC2007_816.
Buffy_109.jpg  Buffy_442.jpg    Inria_28.jpg    Poselet_107.jpg Poselet_9.jpg    VOC2007_150.jpg   VOC2007_484.jpg   VOC2007_817.
Buffy_11.jpg   Buffy_443.jpg    Inria_280.jpg   Poselet_108.jpg Poselet_90.jpg   VOC2007_151.jpg   VOC2007_485.jpg   VOC2007_818.
Buffy_110.jpg  Buffy_444.jpg    Inria_281.jpg   Poselet_109.jpg Poselet_91.jpg   VOC2007_152.jpg   VOC2007_486.jpg   VOC2007_819.
Buffy_111.jpg  Buffy_445.jpg    Inria_282.jpg   Poselet_11.jpg  Poselet_92.jpg   VOC2007_153.jpg   VOC2007_487.jpg   VOC2007_82.
Buffy_112.jpg  Buffy_446.jpg    Inria_283.jpg   Poselet_110.jpg Poselet_93.jpg   VOC2007_154.jpg   VOC2007_488.jpg   VOC2007_820.
Buffy_113.jpg  Buffy_447.jpg    Inria_284.jpg   Poselet_111.jpg Poselet_94.jpg   VOC2007_155.jpg   VOC2007_489.jpg   VOC2007_83.
Buffy_114.jpg  Buffy_448.jpg    Inria_285.jpg   Poselet_112.jpg Poselet_95.jpg   VOC2007_156.jpg   VOC2007_49.jpg    VOC2007_84.
Buffy_115.jpg  Buffy_449.jpg    Inria_286.jpg   Poselet_113.jpg Poselet_96.jpg   VOC2007_157.jpg   VOC2007_490.jpg   VOC2007_85.
Buffy_116.jpg  Buffy_45.jpg     Inria_287.jpg   Poselet_114.jpg Poselet_97.jpg   VOC2007_158.jpg   VOC2007_491.jpg   VOC2007_86.
Buffy_117.jpg  Buffy_450.jpg    Inria_288.jpg   Poselet_115.jpg Poselet_98.jpg   VOC2007_159.jpg   VOC2007_492.jpg   VOC2007_87.
```

- labels目录下的txt，是JPEGImages目录下对应图片所标注的labels，文件名字都是一模一样的，只是文件后缀有不一样。

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/labels$ ls
Buffy_1.txt    Buffy_432.txt    Inria_27.txt    Inria_98.txt    Poselet_8.txt    VOC2007_140.txt    VOC2007_474.txt
Buffy_10.txt   Buffy_433.txt    Inria_270.txt   Inria_99.txt    Poselet_80.txt   VOC2007_141.txt   VOC2007_475.txt
Buffy_100.txt  Buffy_434.txt    Inria_271.txt   Poselet_1.txt   Poselet_81.txt   VOC2007_142.txt   VOC2007_476.txt
Buffy_101.txt  Buffy_435.txt    Inria_272.txt   Poselet_10.txt  Poselet_82.txt   VOC2007_143.txt   VOC2007_477.txt
Buffy_102.txt  Buffy_436.txt    Inria_273.txt   Poselet_100.txt Poselet_83.txt   VOC2007_144.txt   VOC2007_478.txt
Buffy_103.txt  Buffy_437.txt    Inria_274.txt   Poselet_101.txt Poselet_84.txt   VOC2007_145.txt   VOC2007_479.txt
Buffy_104.txt  Buffy_438.txt    Inria_275.txt   Poselet_102.txt Poselet_85.txt   VOC2007_146.txt   VOC2007_48.txt
Buffy_105.txt  Buffy_439.txt    Inria_276.txt   Poselet_103.txt Poselet_86.txt   VOC2007_147.txt   VOC2007_480.txt
Buffy_106.txt  Buffy_44.txt     Inria_277.txt   Poselet_104.txt Poselet_87.txt   VOC2007_148.txt   VOC2007_481.txt
Buffy_107.txt  Buffy_440.txt    Inria_278.txt   Poselet_105.txt Poselet_88.txt   VOC2007_149.txt   VOC2007_482.txt
Buffy_108.txt  Buffy_441.txt    Inria_279.txt   Poselet_106.txt Poselet_89.txt   VOC2007_15.txt    VOC2007_483.txt
Buffy_109.txt  Buffy_442.txt    Inria_28.txt    Poselet_107.txt Poselet_9.txt    VOC2007_150.txt   VOC2007_484.txt
Buffy_11.txt   Buffy_443.txt    Inria_280.txt   Poselet_108.txt Poselet_90.txt   VOC2007_151.txt   VOC2007_485.txt
Buffy_110.txt  Buffy_444.txt    Inria_281.txt   Poselet_109.txt Poselet_91.txt   VOC2007_152.txt   VOC2007_486.txt
Buffy_111.txt  Buffy_445.txt    Inria_282.txt   Poselet_11.txt  Poselet_92.txt   VOC2007_153.txt   VOC2007_487.txt
Buffy_112.txt  Buffy_446.txt    Inria_283.txt   Poselet_110.txt Poselet_93.txt   VOC2007_154.txt   VOC2007_488.txt
Buffy_113.txt  Buffy_447.txt    Inria_284.txt   Poselet_111.txt Poselet_94.txt   VOC2007_155.txt   VOC2007_489.txt
Buffy_114.txt  Buffy_448.txt    Inria_285.txt   Poselet_112.txt Poselet_95.txt   VOC2007_156.txt   VOC2007_49.txt
Buffy_115.txt  Buffy_449.txt    Inria_286.txt   Poselet_113.txt Poselet_96.txt   VOC2007_157.txt   VOC2007_490.txt
Buffy_116.txt  Buffy_45.txt     Inria_287.txt   Poselet_114.txt Poselet_97.txt   VOC2007_158.txt   VOC2007_491.txt
Buffy_117.txt  Buffy_450.txt    Inria_288.txt   Poselet_115.txt Poselet_98.txt   VOC2007_159.txt   VOC2007_492.txt
```

- list目录下的hand_train.txt文件中保存的是JPEGImages目录下所有图片的绝对路径，**注意：如果此处的绝对路径不对，后面模型训练的时候会因为找不到训练的图片，而导致训练失败，所以请一定要把test_dataset/list/hand_test.txt和training_dataset/list/hand_train.txt中的图片绝对路径修改正确。**

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$ ls
hand_test.txt
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_1.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_10.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_100.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_101.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_102.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_103.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_104.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_105.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_106.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_107.jpg
/home/nict02-s32/ai/sample/hand_detect/test_dataset/JPEGImages/VOC2007_108.jpg
```

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/list$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/list$ ls
hand_train.txt
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/list$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/training_dataset/list$
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_1.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_10.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_100.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_101.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_102.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_103.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_104.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_105.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_106.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_107.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_108.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_109.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_11.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_110.jpg
/home/nict02-s32/ai/sample/hand_detect/training_dataset/JPEGImages/Bufgy_111.jpg
```

- 注意：所有的txt文件都必须是linux格式的，可以使用 dos2unix 工具进行文件格式的转换，使用方法：dos2unix filename （如果还不知道如何使用，可上网咨询度娘）

```
sudo apt-get install dos2unix # 如果服务器没有dos2unix这个软件的话，执行这条命令，先进行安装
```

```
dos2unix hand_test.txt # 修改某个文件的文件格式
```

```
dos2unix * # 修改该目录下的所有文件的文件格式
```

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$ dos2unix hand_test.txt
-bash: dos2unix: 未找到命令
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$ sudo apt-get install dos2unix
[sudo] password for nict02-s32:
Reading package lists ... Done
Building dependency tree
Reading state information ... Done
The following NEW packages will be installed:
  dos2unix
0 upgraded, 1 newly installed, 0 to remove and 47 not upgraded.
Need to get 351 kB of archives.
After this operation, 1267 kB of additional disk space will be used.
Get:1 https://repo.huaweicloud.com/ubuntu bionic/universe amd64 dos2unix amd64 7.3.4-3 [351 kB]
Fetched 351 kB in 0s (3546 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
Selecting previously unselected package dos2unix.
(Reading database ... 73722 files and directories currently installed.)
Preparing to unpack .../dos2unix_7.3.4-3_amd64.deb ...
Unpacking dos2unix (7.3.4-3) ...
Setting up dos2unix (7.3.4-3) ...
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$ dos2unix hand_test.txt
dos2unix: converting file hand_test.txt to Unix format ...
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/list$
```

dos2unix工具找不到，则执行下面的命令进行安装

使用dos2unix工具进行格式转换

```
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/labels$
nict02-s32@4a43f76a9207:~/ai/sample/hand_detect/test_dataset/labels$ dos2unix *
dos2unix: converting file VOC2007_100.txt to Unix format ...
dos2unix: converting file VOC2007_101.txt to Unix format ...
dos2unix: converting file VOC2007_102.txt to Unix format ...
dos2unix: converting file VOC2007_103.txt to Unix format ...
dos2unix: converting file VOC2007_104.txt to Unix format ...
dos2unix: converting file VOC2007_105.txt to Unix format ...
dos2unix: converting file VOC2007_106.txt to Unix format ...
dos2unix: converting file VOC2007_107.txt to Unix format ...
dos2unix: converting file VOC2007_108.txt to Unix format ...
dos2unix: converting file VOC2007_109.txt to Unix format ...
dos2unix: converting file VOC2007_10.txt to Unix format ...
dos2unix: converting file VOC2007_110.txt to Unix format ...
dos2unix: converting file VOC2007_111.txt to Unix format ...
dos2unix: converting file VOC2007_112.txt to Unix format ...
dos2unix: converting file VOC2007_113.txt to Unix format ...
dos2unix: converting file VOC2007_114.txt to Unix format ...
dos2unix: converting file VOC2007_115.txt to Unix format ...
dos2unix: converting file VOC2007_116.txt to Unix format ...
dos2unix: converting file VOC2007_117.txt to Unix format ...
```


步骤3：检查cuda环境

- 查看是否能获取cuda,首先在训练服务器的命令行终端输入 python,然后输入import torch, 再输入 torch.cuda.is_available(), 若显示true 则表示cuda 可获取, 否则检查环境是否正确, 退出输入 exit()即可

```
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/dataset$ python
Python 3.6.5 |Anaconda, Inc.| (default, Apr 29 2018, 16:14:56)
[GCC 7.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>> import torch
>>>
>>> torch.cuda.is_available()
True
>>>
>>> exit() 退出 python环境
nict02-s32@4a43f76a9207:~/ai/sample/trash_classify/dataset$
```

步骤4：修改源码和配置文件

- 修改配置文件hand.data中的参数, 注意: hand.data 文件在 ~/ai/framework/darknet-master 目录下

```
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ pwd
/home/nict02-s32/ai/framework/darknet-master
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ ls
LICENSE      LICENSE.gen  LICENSE.meta LICENSE.v1  README.md  cfg      data      hand.data  include
LICENSE.fuck LICENSE.gpl  LICENSE.mit  Makefile   backup     darknet  examples  hand.names libdarknet.
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ vim hand.data
```

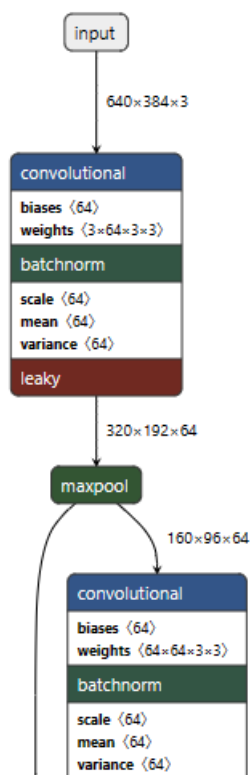
```
classes=1
train =/home/nict02-s32/ai/sample/hand_detect/training_dataset/list/hand_train.txt
valid =/home/nict02-s32/ai/sample/hand_detect/test_dataset/list/hand_test.txt
names =/home/nict02-s32/ai/framework/darknet-master/hand.names
backup =/home/nict02-s32/ai/framework/darknet-master/backup
~
```

classes: 为指定类别数量, 手部检测模型为1, 其他模型需根据实际场景进行填写
train: 指定训练数据list的路径
valid: 指定测试数据list的路径
names: 指定类别的名字
backup: 指定训练后权重的保存路径

- 修改~/ai/framework/darknet-master目录下的hand.name, hand.name里面的内容, 就是需要检测的类型。本文需要检测的物体只有手, 因此hand.name里面的内容就只有hand, 如果您需要检测多种物体, 可每一行书写一种物体。

```
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ cat hand.names
hand
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$
```

- 注意: ~/ai/framework/darknet-master/cfg/下的 resnet18.cfg, 可以通过 <https://netron.app/> 查看网络结构和参数, 部分网络结构截图如下图所示:



步骤5：查看显卡资源

- 使命令 `nvidia-smi` 查看哪张显卡空闲或 Volatile GPU-Util 利用率较低，假如第1张显卡空闲，我们就进入到下一步骤。

```
Fri Feb 10 18:51:08 2023 $ nvidia-smi
```

NVIDIA-SMI 515.48.07 Driver Version: 515.48.07 CUDA Version: 11.7									
GPU	Name	Persistence-M	Bus-Id	Disp.A	Memory-Usage	Volatile GPU-Util	Uncorr. Compute	ECC MIG	
Fan	Temp	Perf	Pwr:Usage/Cap						M. M.
0	Tesla T4	Off	00000000:86:00.0	Off	1371MiB / 15360MiB	0%	Default	0	N/A
N/A	35C	P0	28W / 70W						
1	Tesla T4	Off	00000000:87:00.0	Off	2MiB / 15360MiB	0%	Default	0	N/A
N/A	38C	P0	27W / 70W						
2	Tesla T4	Off	00000000:AF:00.0	Off	2MiB / 15360MiB	0%	Default	0	N/A
N/A	36C	P0	27W / 70W						
3	Tesla T4	Off	00000000:D8:00.0	Off	2MiB / 15360MiB	0%	Default	0	N/A
N/A	38C	P0	28W / 70W						

步骤6：开始训练

- 切换到darknet-master目录下，在训练服务器的命令行终端，执行下面的命令，开始进行模型的训练

```
cd ~/ai/framework/darknet-master
```

```
mkdir backup
```

```
CUDA_VISIBLE_DEVICES=1 ./darknet detector train hand.data cfg/resnet18.cfg
```

对上述命令阐述如下：

`CUDA_VISIBLE_DEVICES=1` 表示使用GPU编号为1的显卡进行训练

`darknet`为可执行文件

`detector`为必选参数，直接沿用即可

`train`指定当前模式为训练模式，测试时候需要改成`test`或者`valid`。

`hand.data`是一个文件，文件中指定了训练中数据、类别、模型存储路径等信息。

`resnet18.cfg`是一个文件，文件中指定了训练的模型。

```
nict02-s32@4a43f76a9207:~$ cd ~/ai/framework/darknet-master
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ mkdir backup
nict02-s32@4a43f76a9207:~/ai/framework/darknet-master$ CUDA_VISIBLE_DEVICES=1 ./darknet detector train hand.data cfg/resnet18.cfg
resnet18
layer   filters  size  input              output             BFLOPs
0 conv   64        3 x 3 / 2  640 x 384 x 3    → 320 x 192 x 64    0.212 BFLOPs
1 max    2 x 2 / 2  320 x 192 x 64  → 160 x 96 x 64
2 conv   64        3 x 3 / 1  160 x 96 x 64    → 160 x 96 x 64    1.132 BFLOPs
3 conv   64        3 x 3 / 1  160 x 96 x 64    → 160 x 96 x 64    1.132 BFLOPs
4 res    1         160 x 96 x 64  → 160 x 96 x 64
5 conv   64        3 x 3 / 1  160 x 96 x 64    → 160 x 96 x 64    1.132 BFLOPs
6 conv   64        3 x 3 / 1  160 x 96 x 64    → 160 x 96 x 64    1.132 BFLOPs
7 res    4         160 x 96 x 64  → 160 x 96 x 64
8 max    2 x 2 / 2  160 x 96 x 64    → 80 x 48 x 64
9 conv   128       1 x 1 / 1  80 x 48 x 64     → 80 x 48 x 128    0.063 BFLOPs
10 conv  128       3 x 3 / 1  80 x 48 x 128    → 80 x 48 x 128    1.132 BFLOPs
11 conv  128       1 x 1 / 1  80 x 48 x 128    → 80 x 48 x 128    0.126 BFLOPs
12 res    9         80 x 48 x 128  → 80 x 48 x 128
13 conv  128       1 x 1 / 1  80 x 48 x 128    → 80 x 48 x 128    0.126 BFLOPs
14 conv  128       3 x 3 / 1  80 x 48 x 128    → 80 x 48 x 128    1.132 BFLOPs
15 conv  128       1 x 1 / 1  80 x 48 x 128    → 80 x 48 x 128    0.126 BFLOPs
16 res    13        80 x 48 x 128  → 80 x 48 x 128
17 max    2 x 2 / 2  80 x 48 x 128    → 40 x 24 x 128
18 conv  256       1 x 1 / 1  40 x 24 x 128    → 40 x 24 x 256    0.063 BFLOPs
19 conv  256       3 x 3 / 1  40 x 24 x 256    → 40 x 24 x 256    1.132 BFLOPs
```

- 训练过程如下图所示：

```
Loaded: 0.000038 seconds
Region Avg IOU: 0.886993, Class: 1.000000, Obj: 0.858286, No Obj: 0.002914, Avg Recall: 1.000000, count: 12
Region Avg IOU: 0.832719, Class: 0.999996, Obj: 0.854385, No Obj: 0.002407, Avg Recall: 1.000000, count: 10
Region Avg IOU: 0.828295, Class: 1.000000, Obj: 0.823431, No Obj: 0.003020, Avg Recall: 1.000000, count: 14
Region Avg IOU: 0.767622, Class: 0.999999, Obj: 0.747484, No Obj: 0.001963, Avg Recall: 1.000000, count: 9
Region Avg IOU: 0.783933, Class: 1.000000, Obj: 0.679804, No Obj: 0.002955, Avg Recall: 1.000000, count: 14
Region Avg IOU: 0.821345, Class: 0.999999, Obj: 0.825026, No Obj: 0.002427, Avg Recall: 1.000000, count: 8
Region Avg IOU: 0.823367, Class: 0.923077, Obj: 0.831396, No Obj: 0.002912, Avg Recall: 1.000000, count: 13
Region Avg IOU: 0.763027, Class: 0.999998, Obj: 0.745521, No Obj: 0.003229, Avg Recall: 0.937500, count: 16
165648: 1.626298, 1.757994 avg, 0.001000 rate, 0.348999 seconds, 7951104 images
Loaded: 0.452634 seconds
Region Avg IOU: 0.808926, Class: 1.000000, Obj: 0.687957, No Obj: 0.002875, Avg Recall: 1.000000, count: 17
Region Avg IOU: 0.806398, Class: 1.000000, Obj: 0.748294, No Obj: 0.003240, Avg Recall: 1.000000, count: 15
Region Avg IOU: 0.666447, Class: 0.937500, Obj: 0.678420, No Obj: 0.002468, Avg Recall: 0.812500, count: 16
Region Avg IOU: 0.795317, Class: 0.999999, Obj: 0.755479, No Obj: 0.002694, Avg Recall: 0.846154, count: 13
Region Avg IOU: 0.646040, Class: 0.999997, Obj: 0.585359, No Obj: 0.003021, Avg Recall: 0.777778, count: 24
Region Avg IOU: 0.771670, Class: 0.999845, Obj: 0.727320, No Obj: 0.002294, Avg Recall: 1.000000, count: 11
Region Avg IOU: 0.861634, Class: 1.000000, Obj: 0.851164, No Obj: 0.003151, Avg Recall: 1.000000, count: 13
Region Avg IOU: 0.759751, Class: 1.000000, Obj: 0.729890, No Obj: 0.003590, Avg Recall: 0.950000, count: 20
165649: 2.393668, 1.821562 avg, 0.001000 rate, 0.345652 seconds, 7951152 images
Loaded: 0.306501 seconds
Region Avg IOU: 0.733155, Class: 1.000000, Obj: 0.751796, No Obj: 0.003858, Avg Recall: 0.882353, count: 17
Region Avg IOU: 0.602319, Class: 0.928571, Obj: 0.637364, No Obj: 0.002698, Avg Recall: 0.642857, count: 14
Region Avg IOU: 0.842528, Class: 0.937500, Obj: 0.852856, No Obj: 0.003894, Avg Recall: 1.000000, count: 16
Region Avg IOU: 0.702595, Class: 0.888887, Obj: 0.663856, No Obj: 0.003068, Avg Recall: 0.777778, count: 18
Region Avg IOU: 0.592232, Class: 0.849999, Obj: 0.551057, No Obj: 0.002706, Avg Recall: 0.750000, count: 20
Region Avg IOU: 0.818521, Class: 1.000000, Obj: 0.752892, No Obj: 0.002339, Avg Recall: 1.000000, count: 9
Region Avg IOU: 0.739972, Class: 1.000000, Obj: 0.744841, No Obj: 0.001861, Avg Recall: 1.000000, count: 9
Region Avg IOU: 0.822828, Class: 0.999999, Obj: 0.798817, No Obj: 0.003436, Avg Recall: 1.000000, count: 16
165650: 1.803199, 1.819725 avg, 0.001000 rate, 0.343814 seconds, 7951200 images
Loaded: 0.224971 seconds
Region Avg IOU: 0.767470, Class: 0.999999, Obj: 0.764907, No Obj: 0.003134, Avg Recall: 0.928571, count: 14
Region Avg IOU: 0.775499, Class: 0.923077, Obj: 0.761286, No Obj: 0.003453, Avg Recall: 0.846154, count: 13
Region Avg IOU: 0.747874, Class: 0.923077, Obj: 0.709428, No Obj: 0.002432, Avg Recall: 0.923077, count: 13
Region Avg IOU: 0.511091, Class: 0.882353, Obj: 0.550108, No Obj: 0.002715, Avg Recall: 0.588235, count: 17
Region Avg IOU: 0.775522, Class: 1.000000, Obj: 0.751023, No Obj: 0.002379, Avg Recall: 1.000000, count: 12
Region Avg IOU: 0.695097, Class: 0.941176, Obj: 0.708488, No Obj: 0.002882, Avg Recall: 0.823529, count: 17
Region Avg IOU: 0.709576, Class: 0.928571, Obj: 0.717968, No Obj: 0.002610, Avg Recall: 0.857143, count: 14
Region Avg IOU: 0.838552, Class: 1.000000, Obj: 0.789735, No Obj: 0.004420, Avg Recall: 1.000000, count: 18
165651: 2.080095, 1.845762 avg, 0.001000 rate, 0.325313 seconds, 7951248 images
Loaded: 0.022999 seconds
```

- 当开始训练之后，我们可以看到显卡0的Volatile GPU-Util 利用率就已经大于0%了，代表该显卡正在使用中。

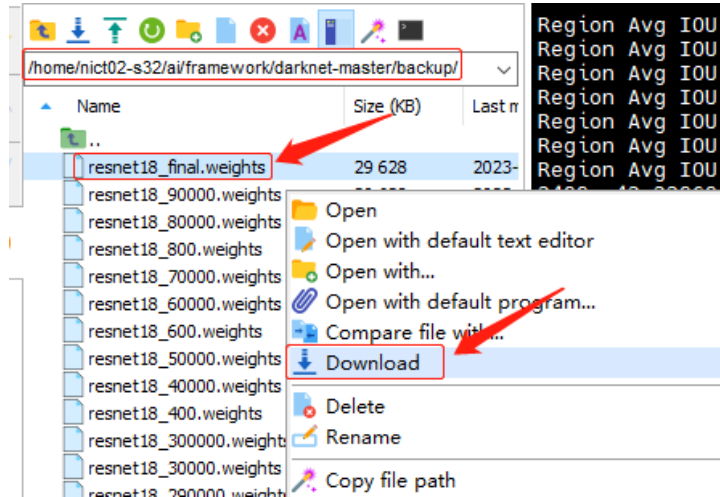

```

resnet18_backup
resnet18_100.weights
resnet18_100000.weights
resnet18_1000000.weights
resnet18_110000.weights
resnet18_120000.weights
resnet18_130000.weights
resnet18_140000.weights
resnet18_150000.weights
resnet18_160000.weights
resnet18_170000.weights
resnet18_180000.weights
resnet18_190000.weights
resnet18_200.weights
resnet18_20000.weights
resnet18_200000.weights
resnet18_210000.weights
resnet18_220000.weights
resnet18_230000.weights
resnet18_240000.weights
resnet18_250000.weights
resnet18_260000.weights
resnet18_270000.weights
resnet18_280000.weights
resnet18_290000.weights
resnet18_300.weights
resnet18_300000.weights
resnet18_3000000.weights
resnet18_400.weights
resnet18_40000.weights
resnet18_500.weights
resnet18_50000.weights
resnet18_600.weights
resnet18_60000.weights
resnet18_700.weights
resnet18_70000.weights
resnet18_800.weights
resnet18_80000.weights
resnet18_900.weights
resnet18_90000.weights
resnet18_final.weights

```

步骤7：模型转换

- 从训练服务器的~/ai/framework/darknet-master/backup 目录下，下载训练好的模型文件 resnet18_final.weights，到windows终端。



- 步骤2：将刚下载到windows的resnet18_final.weights模型文件，通过samba映射的方式上传到你 Ubuntu docker的/root/darknet2caffe/目录下

```

root@hispark:~/darknet2caffe# ls
README.md  __pycache__  caffe_layers  cfg  cfg.py  darknet2caffe.py  prototxt  prototxt.py  resnet18.cfg  resnet18_final.weights  weights
root@hispark:~/darknet2caffe#

```

- 步骤3：执行下面的命令，将darknet模型转换为caffe模型。

```

# 转换命令遵循：
# python cfg[in] weights[in] prototxt[out] caffemodel[out]
# 本文转换命令如下：
python3.8 darknet2caffe.py resnet18.cfg resnet18_final.weights
resnet18.prototxt resnet18.caffemodel

```

如下图所示：

```

root@hispark:~/darknet2caffe#
root@hispark:~/darknet2caffe# python3.8 darknet2caffe.py resnet18.cfg resnet18_final.weights resnet18.prototxt resnet18.caffemodel
WARNING: Logging before InitGoogleLogging() is written to STDERR
0118 20:21:26.820554 81 upgrade_proto.cpp:69] Attempting to upgrade input file specified using deprecated input fields: resnet18.prototxt
0118 20:21:26.820636 81 upgrade_proto.cpp:72] Successfully upgraded file specified using deprecated input fields.
0118 20:21:26.820643 81 upgrade_proto.cpp:74] Note that future Caffe releases will only support input layers and not input fields.
0118 20:21:26.821120 81 net.cpp:53] Initializing net from parameters:
name: "Darknet2Caffe"
state {
  phase: TEST
  level: 0
}
layer {
  name: "input"
  type: "Input"
  top: "data"
  input_param {
    shape {
      dim: 1

```

- 当转换成功后，会在darknet2caffe目录生成一个resnet18.caffemodel和一个resnet18.prototxt文件，如下图所示。

```

0118 20:21:26.984241 81 net.cpp:244] This network produces output layer36-conv
0118 20:21:26.984344 81 net.cpp:257] Network initialization done.
save prototxt to resnet18.prototxt
save caffemodel to resnet18.caffemodel
root@hispark:~/darknet2caffe#
root@hispark:~/darknet2caffe# ls
README.md  __pycache__  caffe_layers  cfg.py  darknet2caffe.py  prototxt  prototxt.py  resnet18.caffemodel  resnet18.prototxt  weights
root@hispark:~/darknet2caffe#

```

电脑 > docker (\\192.168.56.106) (Z:) > root > darknet2caffe				
名称	修改日期	类型	大小	
.git	2022/5/4 17:09	文件夹		
__pycache__	2022/5/4 18:32	文件夹		
caffe_layers	2022/5/4 17:09	文件夹		
cfg	2022/5/4 17:09	文件夹		
prototxt	2022/5/4 17:09	文件夹		
weights	2022/5/4 17:09	文件夹		
cfg.py	2022/5/4 17:09	Python 源文件	11 KB	
darknet2caffe.py	2022/5/4 18:21	Python 源文件	22 KB	
prototxt.py	2022/5/4 18:22	Python 源文件	7 KB	
README.md	2022/5/4 17:09	MD 文件	3 KB	
resnet18.cfg	2022/12/8 9:26	Configuration 源...	4 KB	
resnet18_final.weights	2023/1/18 18:31	WEIGHTS 文件	29,629 KB	
resnet18.prototxt	2023/1/18 20:21	PROTOTXT 文件	19 KB	
resnet18.caffemodel	2023/1/18 20:21	CAFFEMODEL 文...	29,639 KB	

步骤8：模型量化

- 关于模型量化的相关操作，可以直接参考社区提供的 [5.3.2.4.4 模型量化](#) 小节的内容即可。

- 5.3.2.3.8、模型部署板端...
 - 5.3.2.3.8.1、概述
 - 5.3.2.3.8.2、编译
 - 5.3.2.3.8.3、拷贝可执...
 - 5.3.2.3.8.4、拷贝mnt...
 - 5.3.2.3.8.5、功能验证
- 5.3.2.4、检测网
 - 5.3.2.4.1 数据集制作和标注
 - 5.3.2.4.2 本地模型训练
 - 5.3.2.4.3 Darknet模型转...
 - 5.3.2.4.4 模型量化
 - 5.3.2.4.5、模型部署板端...
 - 5.3.2.4.5.1、概述
 - 5.3.2.4.5.2、编译
 - 5.3.2.4.5.3、拷贝可执...
 - 5.3.2.4.5.4、拷贝mnt...

5.3.2.4.4 模型量化

注：nnie_mapper配置概念请阅读源码的device/soc/hisilicon/hi3516dv300/sdk_linux/sample/d南.pdf》3.5.2章节配置文件说明

- 由于yolo2网络的最后一层需要通过AI CPU推理，转换之前，需要手工将其删除，如下图所示

```
layer {
  bottom: "layer36-conv"
  top: "layer36-conv"
  name: "layer36-scale"
  type: "Scale"
  scale_param {
    bias_term: true
  }
}
layer {
  bottom: "layer36-conv"
  top: "layer37-region"
  name: "layer37-region"
```

步骤9：模型部署板端和调试

- 关于模型量化的相关操作，可以直接参考社区提供的 [5.3.2.4.5、模型部署板端和调试](#) 小节的内容即可。

- 5.3.2.3.8.4、拷贝mnt...
- 5.3.2.3.8.5、功能验证
- 5.3.2.4、检测网
 - 5.3.2.4.1 数据集制作和标注
 - 5.3.2.4.2 本地模型训练
 - 5.3.2.4.3 Darknet模型转c...
 - 5.3.2.4.4 模型量化
 - 5.3.2.4.5、模型部署板端...
 - 5.3.2.4.5.1、概述
 - 5.3.2.4.5.2、编译
 - 5.3.2.4.5.3、拷贝可执...
 - 5.3.2.4.5.4、拷贝mnt...

5.3.2.4.5、模型部署板端和调试

5.3.2.4.5.1、概述

- wk模型生成之后，需要将生成的wk文件部署到板端，本文基于SDK sample的yolov2测试，示例代码参考在device/soc/hisilicon/hi3516dv300/sdk_linux/sample/platform/sv sample_nnie.c，如下图所示：

```
BUILD.gn  C sample_nnie.c x
device > soc > hisilicon > hi3516dv300 > sdk_linux > sample > platform > svp > nnie > sample > C samp
2860
2861 /* function: show YOLOV2 sample(image 416x416 U8_C3) */
2862 void SAMPLE_SVP_NNIE_Yolov2(void)
2863 {
2864     const HT_CHAR *pcSrcFile = "/data/nnie_image/rgb_planar/street_cars
```

